Nima Ghourchian

Karaj, Iran

nimaghourchian.github.io|nima.ghourchian01@gmail.com|+989981107809

Education

Islamic Azad University – Karaj Branch *Sep 2021 – Feb 2026*

Bachelor of Mechanical Engineering | GPA: 16.90/20

Relevant coursework: Engineering dynamics: 17/20, Mechanical vibrations: 19.5/20, Automatic control: 19/20, Machine dynamics: 20/20, Engineering mathematics: 18/20, Differential equations: 20/20, Numerical methods: 19/20

Selected Projects

National 3U CubeSat Competition – Qsat *Jan 2024 – Present*

Attitude Determination & Control Team member | Sharif University – Cubisa Team

- Advanced to the final four among 52 teams nationwide.
- Researched and implemented B-dot detumbling and quaternion-based feedback controller for nadir pointing in Simulink.
- Conducted Software-in-the-Loop (SIL) and Processor-in-the-Loop (PIL) testing to verify control algorithms and embedded implementation.

Vision-Based GPS-denied Navigation *Jun 2025 – Jan 2026*

B.Sc. Final Project

- Built a localization pipeline to match simulated UAV imagery with georeferenced satellite images using ResNet and LiteSAM to generate latitude and longitude estimates.
- Designed and implemented a Kalman filter that fuses simulated IMU, barometer and vision-based pose measurements.

Professional Experience

SpaceOmid *Jul 2025 – Sep 2025*

Intern – Omidspac Team

- Participated in and contributed to weekly technical meetings.
- Assisted in sensor calibration, PID tuning of flight controllers and regular test flights.

SpaceOmid *Sep 2025 – Apr 2026*

Flight Control & Simulation Engineer – Omidspac Team

- Collaborated with a team of engineers to test and debug a spraying module prototype.
- Extended the existing manual altitude-control system by integrating an obstacle-responsive module that enabled automatic altitude adjustment during flight.

Technical Skills

Programming Languages: MATLAB, Python, Modern C++
Frameworks & Tools: ROS 2, ArduPilot, PX4, AirSim, Unreal Engine
Software: SolidWorks, STK

Research Interests

Adaptive control	Nonlinear dynamics & control	State estimation
Sensor fusion	Autonomous systems	Multi-agent systems

Languages

Persian: Native **English**: C1 (CEFR)